PV101 Day 7

Chapters 11-12 + guided load assessment and system sizing

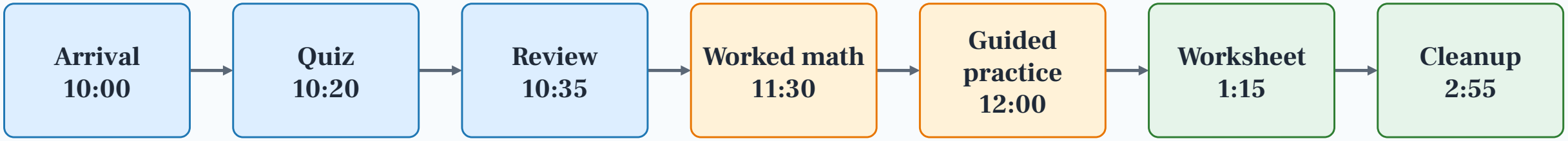
LOADS DRIVE SIZING.

Today is guided class practice. Show the math, units, assumptions, one-line path, and verification checks.

Measure. Verify. Document. Then proceed.

Today's flow

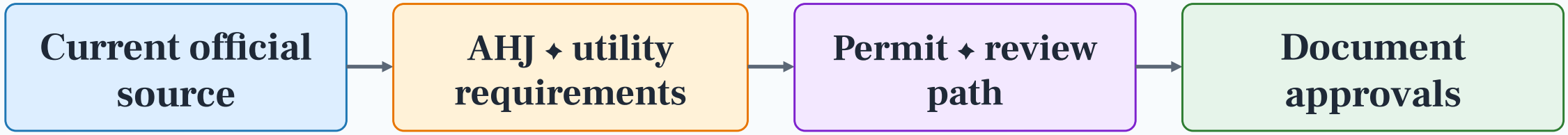
Quiz -> correction -> worked examples -> guided worksheet -> evidence



Morning: finish the reading sequence with Chapters 11-12.
Midday: connect the full PV101 progression to documented loads.
Afternoon: calculate, diagram, peer-check, revise, and obtain signoff.

Chapter 11: work inside local requirements

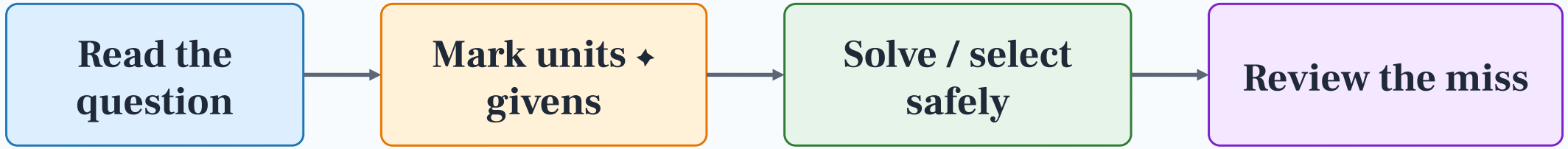
Incentives, resources, permitting, and AHJs



Verify incentives with a current authoritative source; programs and eligibility can change.
Identify the local AHJ and serving utility before assuming the approval process.
Permits and inspections belong to safe, documented work.

Chapter 12: practice with purpose

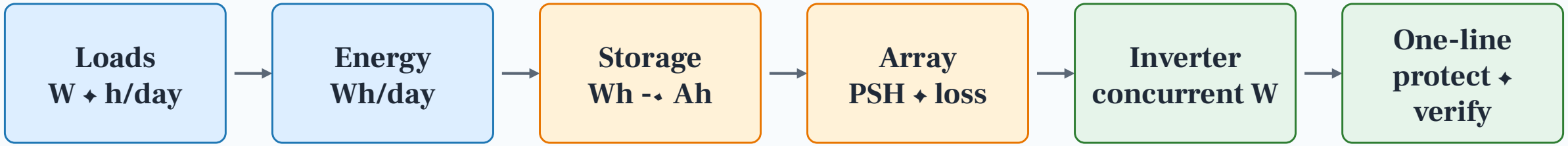
A score is a result. A reviewed mistake is instruction.



Keep units attached to every value.
If two answers appear possible, prioritize safety and documented procedure.
Check whether the result is reasonable before moving on.

The sizing chain

Every later choice starts with documented loads



Do not choose equipment first and invent the load story later.

Worked example: daily energy

Watts x hours = watt-hours

3 lights
3 x 8 W x 4 h
= 96 Wh

Pump
60 W x 0.5 h
= 30 Wh

Vent fan
36 W x 3 h
= 108 Wh

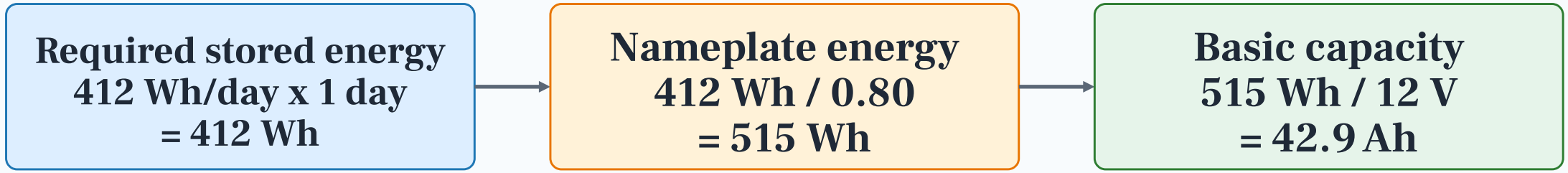
Phone ✦ laptop
48 ✦ 130
= 178 Wh

Daily total = 96 ✦ 30 ✦ 108 ✦ 48 ✦ 130 = 412 Wh/day

Check: every load has power, time, multiplication, and an energy unit.

Worked example: storage

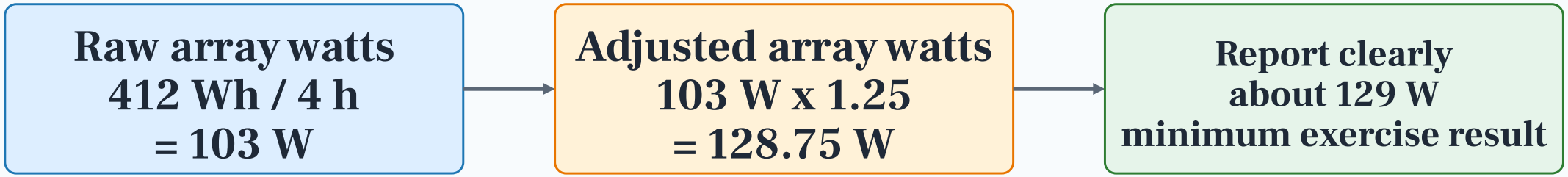
Use the exercise assumptions exactly as documented



Autonomy: 1 day.
Usable fraction / maximum depth of discharge for this exercise: 80%.
System voltage: 12 V.
Round clearly; do not hide the assumption.

Worked example: array watts

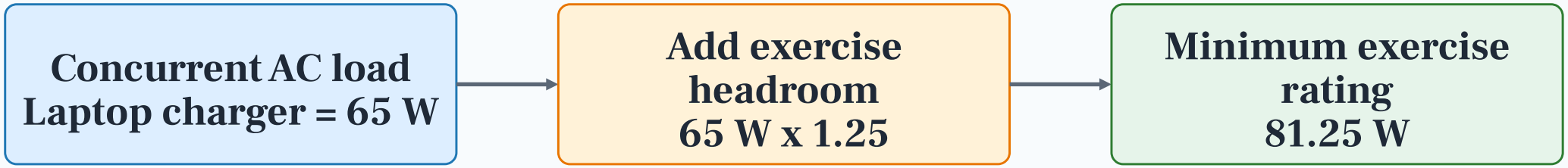
Document peak sun hours and the loss multiplier



Peak sun hours are an input. If the input changes, the answer changes.

Inverter sizing starts with power

Daily energy is not the same as simultaneous load



List the AC loads that can operate at the same time.
Document surge information when a load has a starting surge.
Select from real product ratings only after actual specifications are available.

One-line system path

Show energy flow, return paths, protection, and test points



Mark polarity and the complete return path.
Place protection so it protects the conductor near the source.
Add labels and logical meter points.
Today the diagram is conceptual and must use the provided training assumptions.

Instructor checkpoints

A correct answer must also be inspectable

1	**Loads ✦ hours**	Show units + assumptions
2	**Daily Wh**	Show units + assumptions
3	**Storage Wh/Ah**	Show units + assumptions
4	**Array ✦ inverter**	Show units + assumptions
5	**One-line diagram**	Show units + assumptions
6	**Peer check ✦ signoff**	Revise before signature

Verification check

Before instructor signoff

Loads and hours copied correctly.
Every calculation shows the formula and unit.
Autonomy, usable fraction / depth of discharge, peak sun hours, and loss multiplier match the worksheet.
Answers are reasonable and rounded clearly.
Diagram shows protection, disconnects, labels, meter points, and return path.
Peer check completed and corrections visible.

Day 7 closeout

Reflection, evidence, cleanup, final reminder

Submit
worksheet ✦ reflection

Return
calculators ✦ rulers

Leave
tables clear

Tomorrow: cumulative review and a 25-question final assessment.

Cleanup begins at 2:55. Dismissal follows the room reset.